Python Cheat Sheet

Level 2: keep this beside your keyboard

Types of values

`7`, `-3`	**int**: whole number
`3.5`, `0.1`	**float**: decimal number
`"7"`, `"Nova"`	**str**: text (string)
`True`, `False`	**bool**: yes or no

```
n = int("7") * 2     # text -> int
label = str(42)      # int -> text
print(type(3.5))     # float
```

f-strings: build messages

```
hero = "Nova"
score = 12
print(f"{hero} has {score} points")
print(f"{score / 5:.1f}")  # 2.4
```

and, or, not

```
if score > 10 and lives > 0:
    print("Level up!")
if key == "q" or key == "Escape":
    running = False
if not game_over:
    move()
```

Functions: def and return

```
def area(width, height):
    """Area of a rectangle."""
    return width * height

big = area(3, 4) + 1   # 13
```

`return` hands a value back. `print` only shows it. Variables made inside a function stay inside (scope).

Lists

```
colors = ["orange", "sky", "green"]
first = colors[0]    # index from 0
last = colors[-1]    # the last one
colors.append("blue")  # add at end
print(len(colors))   # 4
for color in colors:
    print(color)
if "green" in colors:
    print("green is there")
pick = random.choice(colors)
```

while loops

```
lives = 3
while lives > 0:
    lives = lives - 1   # must change!

while True:           # ask again
    answer = input("How many? ")
    if answer.isdigit():  # a number?
        break         # leave the loop
    print("Try again")
```

Stuck in a loop that never ends? Press the red **Stop** button.

Nested loops (grids, pixel art)

```
for row in range(3):
    for col in range(4):
        print(row, col)
```

Dictionaries: look up by name

```
player = {"hero": "Nova", "score": 0}
player["score"] = player["score"] + 5
player["lives"] = 3          # new key
for key, value in player.items():
    print(key, value)
if "lives" in player:
    print(player["lives"])
speed = player.get("speed", 1)
```

Strings

```
line = "  12 apples \n"
clean = line.strip()   # "12 apples"
parts = clean.split()  # 2 words
print("Hi".upper(), "Hi".lower())
```

Files: save and load

```
with open("scores.txt", "w") as file:
    file.write("15\n")
with open("scores.txt", "a") as file:
    file.write("9\n")

try:
    with open("scores.txt") as file:
        scores = file.read().split()
except FileNotFoundError:
    scores = []      # no file yet
```

Careful: `"w"` empties the file first!

Testing with pytest

```
def test_area_normal():
    assert area(3, 4) == 12

def test_area_edge():      # edge case
    assert area(0, 4) == 0
```

Test names start with `test_`. To run them, open `my_tests.py` (or `check_my_work.py`) and press Run. No pytest? **Tools > Manage packages**.

Turtle events and games

```
screen = turtle.Screen()
screen.tracer(0)       # no flicker

def go_left():
    player.setx(player.xcor() - 20)

screen.onkey(go_left, "Left")  # no ()
screen.listen()        # hear keys

def game_loop():
    if player.distance(star) < 20:
        print("caught!")   # touching
    screen.update()
    screen.ontimer(game_loop, 50)

game_loop()
turtle.done()
```

Key names: `"Left"`, `"Right"`, `"Up"`, `"Down"`, `"space"`, `"a"`. Click the window first!

Error message decoder

Read the **last line** first. The line above it shows where.

`SyntaxError`	typo: missing `:` `)` or `"`
`IndentationError`	spaces at the start of a line
`NameError`	a name is misspelled or not made yet
`TypeError`	mixing types: `"5" + 5`
`IndexError`	no item at that index (too big)
`KeyError`	that key is not in the dictionary
`FileNotFoundError`	no file with that name in this folder
`ValueError`	`int("hello")` cannot work

Debugging strategies

1. Read the traceback (last line first). **2. Print-debugging:** `print("x is", x)` to see values. **3. Rubber duck:** explain each line out loud.